# ERD to Relational Schema Mapping

Naseej patient health record platform

**Source** docs/diagrams/source/erd-diagram.drawio **Tables** 14 **Relationships** 17

## 1. Mapping rules applied

| ERD construct | Relational rule |
|---|---|
| **Entity (strong)** | Becomes one table. The entity identifier becomes the primary key (PK). |
| **Attribute** | Becomes a column of the table; the data type is kept as drawn in the ERD. |
| **Unique attribute** | Becomes a column with a UNIQUE key (UK), e.g. USERS.email, RECORD_SHARES.token_hash, AUTH_TOKENS.token_hash. |
| **One-to-many (1 : N)** | The primary key of the 1-side is copied into the N-side table as a foreign key (FK). |
| **One-to-one, optional (1 : 0..1)** | As one-to-many, but the foreign key is also declared UNIQUE so at most one row can point to the parent. |
| **Many-to-many (M : N)** | Resolved by an associative table whose composite PK is made of the two foreign keys (SHARE_DOCUMENTS). |
| **Dependent entity** | The table takes the parent key as part of a composite PK (CONTENT_TRANSLATIONS, DOCUMENT_LINKS). |
| **Optional participation (0..1)** | The foreign key column accepts NULL (RECORD_SHARES.prep_id, HEALTH_ENTRIES.source_document_id, HEALTH_ENTRIES.source_fact_id). |

## 2. Relational schema

Notation: **TABLE** ( primary key, *foreign key*, *key that is both*, other columns ). Only the columns shown in the ERD are listed.

**USERS** ( id, email, password_hash, email_verified_at, password_changed_at )

**AUTH_TOKENS** ( id, *user_id*, purpose, token_hash, expires_at, used_at )

**PATIENT_PROFILES** ( id, *user_id*, ai_assistant_consent_at )

**AI_INTERACTIONS** ( id, *patient_id*, kind, outcome )

**CONTENT_TRANSLATIONS** ( *patient_id*, source_hash, target_lang, translated_text )

**HEALTH_ENTRIES** ( id, *patient_id*, kind, title, source, review_status, *source_document_id*, *source_fact_id*, deleted_at )

**SYMPTOM_REPORTS** ( id, *patient_id*, symptom_description, severity, onset_date, captured_via, deleted_at )

**DOCUMENTS** ( id, *patient_id*, file_name, storage_path, doc_type, deleted_at )

**DOCUMENT_FACTS** ( id, *document_id*, patient_decision, extracted_value )

**DOCUMENT_LINKS** ( *document_id*, link_type, target_id, created_at )

**VISIT_PREPS** ( id, *patient_id*, visit_date, status, content )

**RECORD_SHARES** ( id, *patient_id*, *prep_id*, token_hash, snapshot, expires_at, revoked_at )

**SHARE_DOCUMENTS** ( *share_id*, *document_id* )

**SHARE_ACCESS_EVENTS** ( id, *share_id*, event, occurred_at )

## 3. Relationship mapping

Each relationship drawn in the ERD and how it is realised as a foreign key. Cardinality is read parent : child.

| # | Parent | Child | Relationship | Cardinality | Mapping result |
|---|---|---|---|---|---|
| 1 | **USERS** | **PATIENT_PROFILES** | account_for | 1 : 0..1 | FK PATIENT_PROFILES.user_id → USERS.id, with a UNIQUE constraint |
| 2 | **USERS** | **AUTH_TOKENS** | issues | 1 : 0..* | FK AUTH_TOKENS.user_id → USERS.id |
| 3 | **PATIENT_PROFILES** | **AI_INTERACTIONS** | logs_metadata | 1 : 0..* | FK AI_INTERACTIONS.patient_id → PATIENT_PROFILES.id |
| 4 | **PATIENT_PROFILES** | **CONTENT_TRANSLATIONS** | owns | 1 : 0..* | FK is part of the composite PK (patient_id, source_hash, target_lang) |
| 5 | **PATIENT_PROFILES** | **HEALTH_ENTRIES** | owns | 1 : 0..* | FK HEALTH_ENTRIES.patient_id → PATIENT_PROFILES.id |
| 6 | **PATIENT_PROFILES** | **SYMPTOM_REPORTS** | records | 1 : 0..* | FK SYMPTOM_REPORTS.patient_id → PATIENT_PROFILES.id |
| 7 | **PATIENT_PROFILES** | **DOCUMENTS** | owns | 1 : 0..* | FK DOCUMENTS.patient_id → PATIENT_PROFILES.id |
| 8 | **PATIENT_PROFILES** | **VISIT_PREPS** | prepares | 1 : 0..* | FK VISIT_PREPS.patient_id → PATIENT_PROFILES.id |
| 9 | **PATIENT_PROFILES** | **RECORD_SHARES** | creates | 1 : 0..* | FK RECORD_SHARES.patient_id → PATIENT_PROFILES.id |
| 10 | **VISIT_PREPS** | **RECORD_SHARES** | source_prep | 0..1 : 0..* | Optional FK RECORD_SHARES.prep_id → VISIT_PREPS.id (nullable) |
| 11 | **DOCUMENTS** | **DOCUMENT_FACTS** | has_suggestions | 1 : 0..* | FK DOCUMENT_FACTS.document_id → DOCUMENTS.id |
| 12 | **DOCUMENTS** | **HEALTH_ENTRIES** | source_document | 0..1 : 0..* | Optional FK HEALTH_ENTRIES.source_document_id → DOCUMENTS.id (nullable) |
| 13 | **DOCUMENT_FACTS** | **HEALTH_ENTRIES** | confirmed_into | 0..1 : 0..* | Optional FK HEALTH_ENTRIES.source_fact_id → DOCUMENT_FACTS.id (nullable) |
| 14 | **DOCUMENTS** | **DOCUMENT_LINKS** | linked_by | 1 : 0..* | FK is part of the composite PK (document_id, link_type, target_id) |
| 15 | **DOCUMENTS** | **SHARE_DOCUMENTS** | included | 1 : 0..* | M:N resolved: FK is part of the composite PK (share_id, document_id) |
| 16 | **RECORD_SHARES** | **SHARE_DOCUMENTS** | grants | 1 : 0..* | M:N resolved: FK is part of the composite PK (share_id, document_id) |
| 17 | **RECORD_SHARES** | **SHARE_ACCESS_EVENTS** | records | 1 : 0..* | FK SHARE_ACCESS_EVENTS.share_id → RECORD_SHARES.id |

# 4. Table specifications

PK = primary key, FK = foreign key, UK = unique key. Data types are those shown in the ERD.

### USERS

*System account (credentials, verification and password changes)*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| email | text | UK | |
| password_hash | text | | |
| email_verified_at | timestamptz | | |
| password_changed_at | timestamptz | | |

### AUTH_TOKENS

*Single-use email verification and password reset links*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| user_id | uuid | FK | USERS.id |
| purpose | varchar | | |
| token_hash | char | UK | |
| expires_at | timestamptz | | |
| used_at | timestamptz | | |

### PATIENT_PROFILES

*Patient identity; owner of every record below*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| user_id | uuid | FK, UK | USERS.id |
| ai_assistant_consent_at | timestamptz | | |

### AI_INTERACTIONS

*Metadata log of assistant calls*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| patient_id | uuid | FK | PATIENT_PROFILES.id |
| kind | varchar | | |
| outcome | varchar | | |

### CONTENT_TRANSLATIONS

*Cached translation per patient, source and language*

| Column | Data type | Key | References |
|---|---|---|---|
| **patient_id** | uuid | PK, FK | PATIENT_PROFILES.id |
| **source_hash** | char | PK | |
| **target_lang** | char | PK | |
| translated_text | text | | |

## HEALTH_ENTRIES

*Conditions, medications, allergies, measurements*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| patient_id | uuid | FK | PATIENT_PROFILES.id |
| kind | varchar | | |
| title | varchar | | |
| source | varchar | | |
| review_status | varchar | | |
| source_document_id | uuid | FK | DOCUMENTS.id |
| source_fact_id | uuid | FK | DOCUMENT_FACTS.id |
| deleted_at | timestamptz | | |

## SYMPTOM_REPORTS

*Patient-logged symptoms*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| patient_id | uuid | FK | PATIENT_PROFILES.id |
| symptom_description | text | | |
| severity | int | | |
| onset_date | timestamptz | | |
| captured_via | varchar | | |
| deleted_at | timestamptz | | |

## DOCUMENTS

*Uploaded files kept in the patient vault*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| patient_id | uuid | FK | PATIENT_PROFILES.id |
| file_name | varchar | | |
| storage_path | varchar | | |
| doc_type | varchar | | |
| deleted_at | timestamptz | | |

## DOCUMENT_FACTS

*Suggestions extracted from a document*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| document_id | uuid | FK | DOCUMENTS.id |
| patient_decision | varchar | | |
| extracted_value | text | | |

## DOCUMENT_LINKS

*Links a document to a symptom, entry or visit*

| Column | Data type | Key | References |
|---|---|---|---|
| **document_id** | uuid | PK, FK | DOCUMENTS.id |
| **link_type** | varchar | PK | |
| **target_id** | uuid | PK | (polymorphic, see note 3) |
| created_at | timestamptz | | |

## VISIT_PREPS

*Visit preparation sheet*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| patient_id | uuid | FK | PATIENT_PROFILES.id |
| visit_date | date | | |
| status | varchar | | |
| content | jsonb | | |

## RECORD_SHARES

*Time-limited share of approved content*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| patient_id | uuid | FK | PATIENT_PROFILES.id |
| prep_id | uuid | FK | VISIT_PREPS.id |
| token_hash | char | UK | |
| snapshot | jsonb | | |
| expires_at | timestamptz | | |
| revoked_at | timestamptz | | |

## SHARE_DOCUMENTS

*Documents a share may stream*

| Column | Data type | Key | References |
|---|---|---|---|
| **share_id** | uuid | PK, FK | RECORD_SHARES.id |
| **document_id** | uuid | PK, FK | DOCUMENTS.id |

## SHARE_ACCESS_EVENTS

*Append-only access log of a share*

| Column | Data type | Key | References |
|---|---|---|---|
| **id** | uuid | PK | |
| share_id | uuid | FK | RECORD_SHARES.id |
| event | varchar | | |
| occurred_at | timestamptz | | |

## 5. Mapping notes

1. **Ownership.** Every record table reaches PATIENT_PROFILES either directly through patient_id or through its parent (DOCUMENT_FACTS and DOCUMENT_LINKS through DOCUMENTS; SHARE_DOCUMENTS and SHARE_ACCESS_EVENTS through RECORD_SHARES). USERS and AUTH_TOKENS belong to the account, not to the profile.
2. **Provenance foreign keys.** HEALTH_ENTRIES.source_document_id and HEALTH_ENTRIES.source_fact_id reference DOCUMENTS and DOCUMENT_FACTS. They record where an entry came from, are optional (NULL for an entry the patient typed), and are drawn as the source_document and confirmed_into relationships.
3. **Polymorphic key.** DOCUMENT_LINKS.target_id points to a symptom report, a health entry or a visit depending on link_type. A single database foreign key cannot express this, so it is part of the PK without an FK constraint and the application checks it.
4. **Profile to account.** PATIENT_PROFILES.user_id is UNIQUE because the ERD allows at most one profile per account (1 : 0..1).
5. **Soft delete.** HEALTH_ENTRIES, SYMPTOM_REPORTS and DOCUMENTS keep deleted_at instead of removing rows.
6. **Auth tokens.** AUTH_TOKENS (migration 007) holds the links mailed for email verification and password reset. Only the SHA-256 of a token is stored (token_hash, UNIQUE); a token is single-use (used_at) and expires (expires_at). USERS.password_changed_at invalidates sessions issued before a reset. Rows are deleted with their user (ON DELETE CASCADE).